

The Pancreas

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The adult pancreas is a transversely oriented retroperitoneal organ extending from the C-loop of the duodenum to the hilum of the spleen (Fig. 19.1A). Although the organ gets its name from the Greek *pankreas* (“all flesh”), it is, in fact, a complex lobulated organ with distinct exocrine and endocrine components.

The *exocrine pancreas* constitutes 80% to 85% of the organ and is composed of acinar cells. These pyramidally shaped epithelial cells contain membrane-bound granules rich in proenzymes (zymogens), including trypsinogen, chymotrypsinogen, procarboxypeptidase, proelastase, kallikreinogen, and phospholipase A and B. Upon secretion, these proenzymes and enzymes are carried by a series of ductules and ducts to the duodenum, where they are activated by proteolytic cleavage (described later) to promote digestion.

The *endocrine pancreas* is composed of about 1 million endocrine cell clusters, the islets of Langerhans, that are scattered throughout the gland. Islet cells secrete insulin, glucagon, somatostatin, and pancreatic polypeptide. Although they constitute only 1% to 2% of the organ mass, the hormones released by the islet cells are essential regulators of systemic metabolism, particularly glucose metabolism. Diseases of the endocrine pancreas are described in detail in Chapter 24.

CONGENITAL ANOMALIES

The complex process by which the dorsal and ventral pancreatic primordia fuse (between the fifth and eighth gestational weeks) during pancreatic development is imperfect, giving rise to variations in pancreatic anatomy. The pancreas normally arises from the fusion of dorsal and ventral outpouchings of the foregut. The body, tail, and the superior/anterior aspect of the head of the pancreas, as well as the accessory duct of Santorini, are derived from the dorsal primordium. Typically, the ventral primordium gives rise to the posterior/inferior part of the head of the pancreas and drains through the main pancreatic duct into the papilla of Vater.

Pancreas Divisum

Pancreas divisum is the most common congenital anomaly of the pancreas with an incidence of 3% to 10%. In most individuals, the main pancreatic duct (the duct of Wirsung) joins the common bile duct just proximal to the papilla of Vater, and the accessory pancreatic duct (the duct of Santorini) drains into the duodenum through a separate minor papilla (see Fig. 19.1A). Pancreas divisum is caused by a failure of fusion of the fetal duct systems of the dorsal and ventral pancreatic primordia. As a result, in this anatomic variant the bulk of the pancreas (formed by the dorsal pancreatic primordium) drains into the duodenum through the small-caliber minor papilla (Fig. 19.1B), while the duct of Wirsung drains only a small portion of the head of the gland through the papilla of Vater. Most of the patients with pancreas divisum are asymptomatic, and only 5% develop pancreatitis. It has been suggested that inadequate drainage of the pancreatic secretions through the minor papilla, especially when combined with genetic defects that also increase susceptibility to pancreatitis (described later), may predispose individuals with pancreatic divisum to chronic pancreatitis.

Annular Pancreas

An annular pancreas, resulting from incomplete rotation of ventral pancreatic bud, is a band-like ring of normal pancreatic tissue that completely encircles the second portion of the duodenum. It can be an incidental finding on imaging. Approximately one-third cases are asymptomatic. In a subset of cases, an annular pancreas can produce duodenal obstruction.

Ectopic Pancreas

Pancreatic tissue that is aberrantly situated, or ectopic, is found in about 2% of careful routine postmortem examinations. The favored sites for ectopia are the stomach and duodenum, followed by the jejunum, Meckel diverticula, and ileum. Although usually incidental findings, these embryologic rests, composed of normal-appearing pancreatic acini, glands, and sometimes islets of Langerhans, may cause pain from localized inflammation or may incite mucosal bleeding.

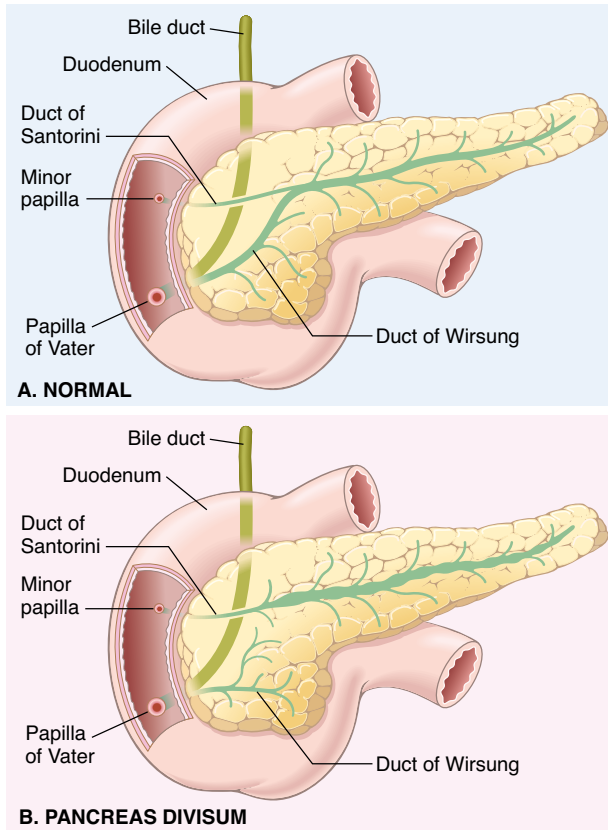


Fig. 19.1 Pancreatic ductal anatomy. (A) Normal ductal anatomy. (B) Ductal anatomy in pancreatic divisum. (Modified from Gregg JA, Monaco AP, McDermott WV. Pancreas divisum: results of surgical intervention. *Am J Surg.* 1983;145:488-492.)

Agenesis

Very rarely the pancreas fails to develop (agenesis). Some cases of agenesis are caused by homozygous germline mutations involving *PDX1*, a gene encoding a homeobox transcription factor that is critical for pancreatic development.

PANCREATITIS

Pancreatitis may be acute or chronic, each with its own characteristic pathologic and clinical features; both are initiated by injuries that lead to autodigestion of the pancreas by its own enzymes. Under normal circumstances, several factors protect the pancreas from autodigestion.

- Most digestive enzymes are synthesized as inactive proenzymes (zymogens) and packaged within secretory granules.
- Proenzymes are typically activated by trypsin, which itself is activated by duodenal enteropeptidase (enterokinase) in the small bowel; as a result, intrapancreatic activation of proenzymes is normally minimal.
- Acinar and ductal cells secrete trypsin inhibitors, including serine protease inhibitor Kazal type I (SPINK1), which further limit intrapancreatic trypsin activity.

Pancreatitis occurs when these protective mechanisms are disrupted or overwhelmed. Acute attacks may range from mild and self-limited to life-threatening events. Recurrent or persistent pancreatitis may lead to permanent loss of pancreatic function or a chronic condition with some residual function.

TABLE 19.1 Etiologic Factors in Acute Pancreatitis

Metabolic
Alcoholism ^a
Hyperlipoproteinemia
Hypercalcemia
Drugs (e.g., azathioprine, statins, GLP-1 agonists, DPP-4 inhibitors)
Genetic
Mutations in genes encoding trypsin (<i>PRSS1</i>), trypsin regulators (<i>SPINK1</i>), or proteins like calcium-sensing receptor that regulate calcium metabolism (<i>CASR</i>)
Cystic fibrosis (<i>CFTR</i>)
Mechanical
Gallstones ^a
Trauma
Iatrogenic injury
Operative injury
Endoscopic procedures with dye injection (e.g., ERCP)
Vascular
Shock
Atheroembolism
Vasculitis (e.g., polyarteritis nodosa)
Infectious
Mumps
Coxsackievirus

^aMost common etiologies in the United States.

DPP-4, Dipeptidyl peptidase-4; ERCP, endoscopic retrograde cholangiopancreatography; GLP-1, glucagon-like peptide-1.

Acute Pancreatitis

Acute pancreatitis is characterized by reversible pancreatic parenchymal injury and inflammation and has many causes, including toxic exposures (e.g., alcohol), pancreatic duct obstruction (e.g., biliary calculi), inherited genetic defects, vascular injury, and infections. Acute pancreatitis is relatively common with an annual incidence in countries with high alcohol consumption of 4.9 to 35 cases per 100,000 people. Biliary tract disease and alcoholism account for approximately 80% of these cases (Table 19.1). The proportion of cases caused by excessive alcohol intake varies from 65% in the United States to 20% in Sweden to 5% or less in southern France and the United Kingdom. The male-to-female ratio is 6:1 in those with alcoholism and 1:3 in patients with biliary tract disease. Gallstones are present in 35% to 60% of acute pancreatitis cases and cause “gallstone pancreatitis” in about 5% of patients with gallstones.

Pathogenesis

Acute pancreatitis results from inappropriate release and activation of pancreatic enzymes that, in turn, destroy pancreatic tissue and elicit an acute inflammatory reaction. Aberrant trypsin activation within the pancreas can activate other proenzymes such as phospholipase and proelastase, which degrade fat cells and damage the elastic fibers of blood vessels, respectively. Once tissue damage is initiated, trypsin can also directly or indirectly activate factors found in the blood, including components of the coagulation, complement, kallikrein, and fibrinolytic pathways (Chapters 3 and 4). The resulting inflammation and small-vessel thrombosis result in further damage to acinar cells, amplifying intrapancreatic enzyme activation.

The triggers that cause intrapancreatic enzyme activation in sporadic acute pancreatitis remain a topic of investigation, but there is evidence for at least three major initiating events (Fig. 19.2):

- **Pancreatic duct obstruction** is most often caused by gallstones and biliary sludge but can also stem from periampullary neoplasms

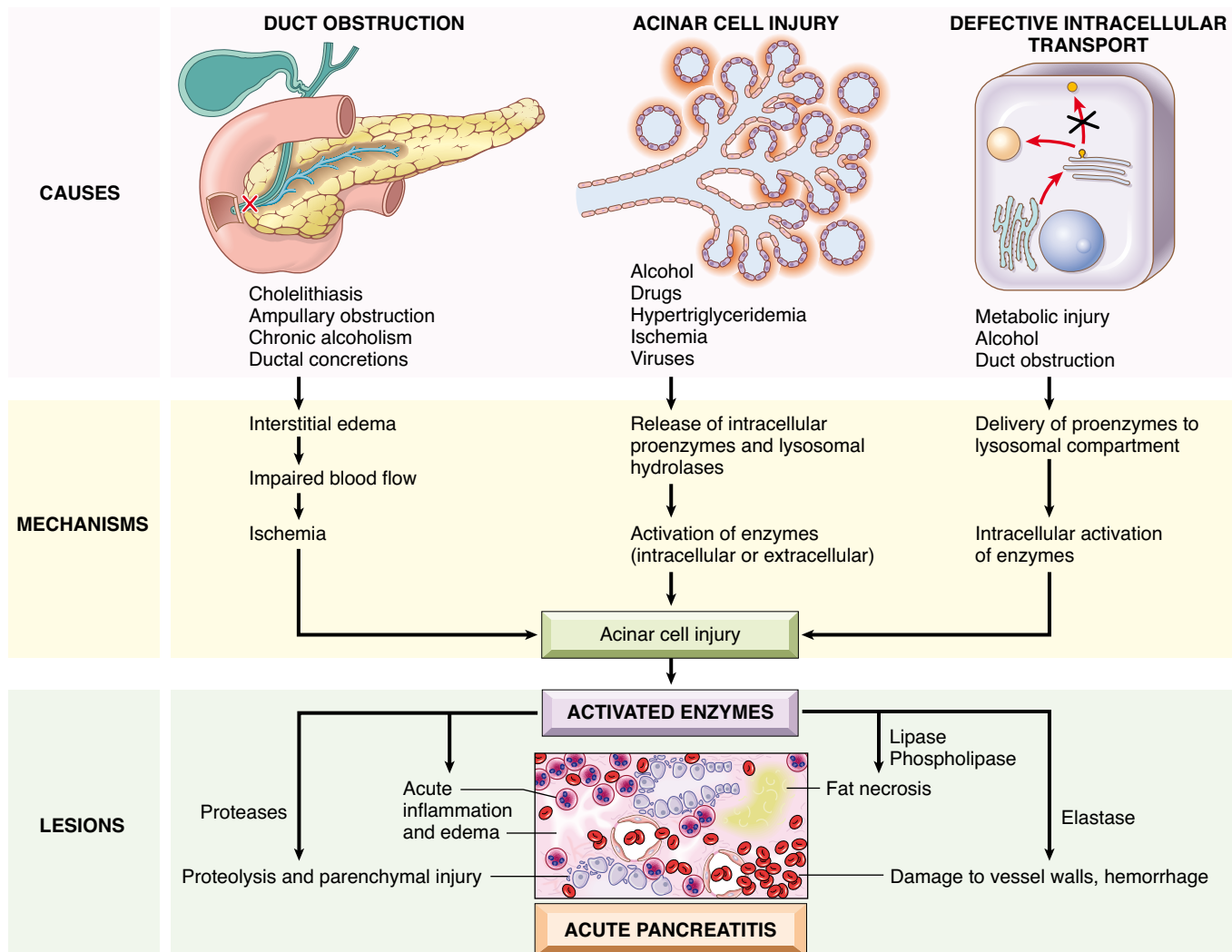


Fig. 19.2 Three proposed pathways in the pathogenesis of acute pancreatitis. Although often depicted as distinct, these pathways may be interconnected (e.g., duct obstruction).

(e.g., pancreatic cancer), choledochoceles (congenital cystic dilation of the common bile duct), parasites (particularly *Ascaris lumbricoides* and *Clonorchis sinensis*), and pancreas divisum. Obstruction raises intrapancreatic ductal pressure and leads to the leakage of enzyme-rich fluid within the interstitium. Once within the interstitium, pancreatic enzymes like lipase, which is secreted in an active form, can lead to local fat necrosis, initiating a vicious cycle. The death of adipocytes is hypothesized to produce “danger” signals locally that are sensed by pancreatic stellate cells and leukocytes, which release proinflammatory mediators that promote microvascular leak and the development of interstitial edema. Edema may further compromise local blood flow, causing vascular insufficiency and ischemic injury to acinar cells.

- **Primary acinar cell injury** leads to the release of digestive enzymes, inflammation, and tissue autodigestion. Acinar cells can be damaged by a variety of endogenous (e.g., oxidative stress), exogenous (e.g., trauma), and iatrogenic (e.g., surgery) insults. Many of these result in oxidative stress and the generation of intracellular free radicals that lead to membrane lipid oxidation and transcription factor activation. The latter include activator protein-1 (AP-1) and nuclear factor κ B (NF- κ B), which induce the expression of chemokines that attract mononuclear cells. Increased calcium flux is another important trigger

for inappropriate enzyme activation. Calcium has a key role in trypsin regulation. When calcium levels are low, trypsin tends to cleave and inactivate itself, but this autoinhibition is abrogated and trypsin autoactivation is favored when calcium levels are increased. Any factor causing increased calcium within acinar cells may therefore trigger excessive trypsin activation. Examples include inherited abnormalities that affect calcium levels (Table 19.2).

- **Defective intracellular transport of proenzymes within acinar cells.** In normal acinar cells, digestive enzymes and lysosomal hydrolases are transported via separate pathways. In animal models of acinar injury, pancreatic proenzymes, of which the cathepsin family of hydrolases is most important, are aberrantly delivered to lysosomes. The hydrolases then activate pancreatic proenzymes that disrupt lysosomal membranes, ultimately leading to release of activated enzymes. It is not clear if this mechanism is relevant to human acute pancreatitis.

Two factors have been proposed to contribute to alcohol-induced pancreatitis: duct obstruction and acinar cell damage. Alcohol consumption transiently increases contraction of the sphincter of Oddi (the muscle at the papilla of Vater), and chronic alcohol ingestion results in the secretion of protein-rich pancreatic fluid that is

TABLE 19.2 Mutated Genes Associated With Inherited Predisposition to Pancreatitis

Gene (Effect of Pathogenic Mutations)	Protein Product	Function
<i>CFTR</i> (loss of function)	Cystic fibrosis transmembrane conductance regulator	Epithelial anion channel. Loss-of-function mutations alter fluid pressure and limit bicarbonate secretion, leading to inspissation of secreted fluids and duct obstruction
<i>PRSS1</i> (gain of function)	Serine protease 1 (trypsinogen 1)	Cationic trypsin. Gain-of-function mutations prevent self-inactivation of trypsin
<i>SPINK1</i> (loss of function)	Serine peptidase inhibitor, Kazal type 1	Inhibitor of trypsin; mutations cause loss-of-function, increasing trypsin activity
<i>CASR</i> (loss of function)	Calcium-sensing receptor	Membrane-bound receptor that senses extracellular calcium levels and controls luminal calcium levels; mutations may alter calcium concentrations and activate trypsin
<i>CTRC</i> (loss of function)	Chymotrypsin C (caldecrin)	Degrades trypsin, protects the pancreas from trypsin-related injury
<i>CPA1</i> (toxic gain of function)	Carboxypeptidase A1	Exopeptidase involved in regulating zymogen activation

prone to form inspissated protein plugs that obstruct small pancreatic ducts. Within acinar cells, alcohol-induced oxidative stress may generate free radicals and also promote the fusion of lysosomes and zymogen granules and increase intracellular calcium levels, possibly through mitochondrial damage, thereby promoting the intracellular activation of trypsin and other digestive enzymes. It should be noted that most people with alcoholism never develop pancreatitis and those who do usually do so after many years of excessive alcohol consumption. Thus, key aspects of the pathophysiology of alcohol-induced pancreatitis remain obscure.

Other proven or suspected triggers of acute pancreatitis in sporadic cases include (see Table 19.1):

- **Metabolic disorders:** such as hypertriglyceridemia and hypercalcemic states, as in hyperparathyroidism.
- **Genetic lesions:** described later.
- **Medications:** Hundreds of drugs have been anecdotally linked to acute pancreatitis, but only about 50 have a definite association. The latter include azathioprine, angiotensin-converting enzyme (ACE) inhibitors, statins, and antidiabetic drugs such as glucagon-like peptide-1 (GLP-1) agonists and dipeptidyl peptidase-4 (DPP-4) inhibitors.
- **Acinar cell injury:** traumatic injury by either blunt abdominal trauma or iatrogenic injury during surgery or endoscopic retrograde cholangiopancreatography; and ischemic injury due to shock, vascular thrombosis, embolism, or vasculitis.
- **Infections:** including mumps, can lead to acute pancreatitis through direct acinar cell injury.

Hereditary factors are increasingly recognized as causes of pancreatitis. Affected individuals have recurrent attacks of severe acute pancreatitis, which often begin in childhood, and can ultimately lead to chronic pancreatitis. The feature shared by most forms of hereditary pancreatitis is a defect that increases or sustains the activity of trypsin (see Table 19.2). Three genes implicated in hereditary pancreatitis deserve special note: *PRSS1*, *SPINK1*, and *CFTR*. Most hereditary cases are due to gain-of-function mutations in the trypsinogen gene (known as *PRSS1*). Some of these *PRSS1* gene mutations make trypsin resistant to self-inactivation, abrogating an important negative feedback mechanism; other mutations make trypsinogen more prone to proteolytic activation. Hereditary pancreatitis associated with trypsinogen mutations has an autosomal-dominant mode of inheritance, as is typically true of disorders associated with gain-of-function mutations.

Hereditary pancreatitis can also be caused by loss-of-function mutations in *SPINK1*, a gene encoding a trypsin inhibitor. Because one

functional copy of *SPINK1* produces sufficient inhibitor to maintain adequate trypsin control, this form of hereditary pancreatitis has an autosomal recessive mode of inheritance.

As discussed in detail in Chapter 10, cystic fibrosis is caused by mutations in the cystic fibrosis transmembrane conductance regulator (*CFTR*) gene that encodes an anion channel that transports chloride and bicarbonate ions. Loss of *CFTR* leads to acidic secretions (poor in bicarbonate) that promote mucin precipitation and plugging of ducts. Obstruction of the pancreatic duct leads to the development of pancreatitis. Disease can occur in those with homozygous and even heterozygous *CFTR* gene mutations, particularly in patients who also have *SPINK1* mutations.

Of note, patients with *PRSS1*-associated hereditary pancreatitis have a 40% lifetime risk of developing pancreatic cancer, yet another example of the nefarious association between neoplasia and chronic tissue injury and inflammation.

MORPHOLOGY

The morphology of acute pancreatitis ranges from limited inflammation and edema to extensive necrosis and hemorrhage. The basic alterations are **(1) microvascular leak and edema, (2) fat necrosis, (3) acute inflammation, (4) damage of pancreatic parenchyma (including autodigestion), and (5) blood vessel destruction with interstitial hemorrhage.** The extent of each of these depends on the duration and severity of the disease.

In more limited forms of **acute interstitial pancreatitis**, histologic alterations are limited to mild inflammation, interstitial edema, and focal fat necrosis within the pancreas and the peripancreatic fat (Fig. 19.3). Fat necrosis, triggered by lipase activity, leads to **saponification**, a process in which fatty acids combine with calcium to form insoluble calcium soaps that impart a granular blue microscopic appearance to surviving fat cells (Chapter 2).

In **acute necrotizing pancreatitis**, acini, ducts, and even islets undergo necrosis. In the most severe form, **hemorrhagic pancreatitis**, extensive parenchymal necrosis is accompanied by intraparenchymal hemorrhage due to vascular injury. This imparts a red-black color with interspersed foci of yellow-white, chalky fat necrosis (Fig. 19.4). Fat necrosis may also occur in the omentum and the mesentery of the bowel adjacent to the pancreas and beyond the abdominal cavity, such as in the subcutaneous fat, because of systemic lipase release. The peritoneal cavity typically contains a serous, slightly turbid, brown-tinged fluid with fat globules that reflect digestion of adipose tissue.

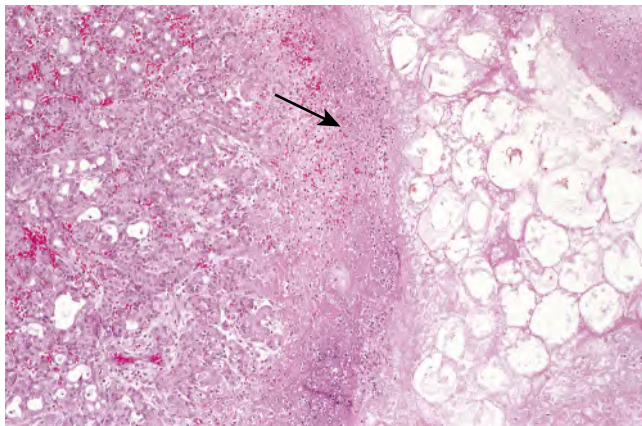


Fig. 19.3 Acute pancreatitis. The microscopic field shows a region of fat necrosis (right) and focal pancreatic parenchymal necrosis (arrow).



Fig. 19.4 Hemorrhagic pancreatitis. The pancreas has been sectioned longitudinally to reveal dark areas of hemorrhage in the head of the pancreas (arrow) and a focal area of pale fat necrosis in the peripancreatic fat (arrowhead).

Clinical Features

Abdominal pain is the cardinal manifestation of acute pancreatitis. Characteristically the pain is constant, intense, and referred to the upper or middle back and, occasionally, the left shoulder. The severity ranges from mild discomfort to incapacitating pain. Anorexia, nausea, and vomiting are common. Elevated plasma levels of amylase and lipase, released from the damaged pancreatic acinar cells, support the diagnosis of acute pancreatitis, as does the exclusion of other causes of abdominal pain.

Severe acute pancreatitis is a medical emergency. Patients usually present with an acute abdomen and systemic findings caused by the release of toxic enzymes, cytokines, and other mediators into the circulation. These activate a systemic inflammatory response, resulting in leukocytosis, disseminated intravascular coagulation, edema, and acute respiratory distress syndrome. Shock, due to the systemic inflammatory response syndrome (Chapter 4), and acute renal tubular necrosis may occur.

Laboratory findings include elevation of serum amylase and lipase levels during the first 4 to 12 hours following the onset of pain. Serum lipase is the most specific and sensitive marker of acute pancreatitis, as serum amylase has a short half-life and may return to normal in 3 to 5 days, whereas lipase levels remain elevated for 8 to 14 days.

Glycosuria occurs in 10% of cases, and hypocalcemia may result from saponification of necrotic fat. Direct visualization of the enlarged inflamed pancreas by computed tomography (CT) scanning is useful in the diagnosis of pancreatitis.

Treatment centers on “resting” the pancreas by total elimination of oral intake and supportive therapy with intravenous fluids and analgesia. Most individuals with acute pancreatitis recover fully but about 5% with severe acute pancreatitis die in the first week of illness. Acute respiratory distress syndrome and acute renal failure are ominous complications. Possible sequelae include sterile pancreatic abscesses and pancreatic pseudocysts. In 40% to 60% of patients with acute necrotizing pancreatitis the acellular debris becomes infected, usually by gram-negative organisms from the gut, further complicating the clinical course. Systemic organ failure and pancreatic necrosis are both adverse prognostic indicators.

Chronic Pancreatitis

Chronic pancreatitis is defined as prolonged inflammation of the pancreas associated with irreversible destruction of exocrine parenchyma, resultant fibrosis, and, in the late stages, loss of endocrine parenchyma. The prevalence of chronic pancreatitis is between 0.04% and 5%; most affected patients are middle-aged males. The most common cause of chronic pancreatitis is long-term alcohol use. In addition to alcohol, chronic pancreatitis has been associated with the following conditions:

- *Obstruction* of the pancreatic duct by calculi or neoplasms
- *Autoimmune injury*
- *Hereditary factors*, as discussed above; up to 25% of chronic pancreatitis has a genetic basis

Pathogenesis

Chronic pancreatitis often follows repeated episodes of acute pancreatitis. It has been proposed that acute pancreatitis initiates a sequence of periductal fibrosis, duct distortion, and altered secretions that, because of recurrent injury, leads to loss of exocrine parenchyma and fibrosis.

Chronic pancreatic injury of any cause leads to local production of inflammatory mediators that promote fibrosis and acinar cell loss. While there is overlap between the cytokines released during chronic and acute pancreatitis, fibrogenic factors tend to predominate in chronic pancreatitis. These fibrogenic cytokines, including transforming growth factor- β (TGF- β) and platelet-derived growth factor (PDGF), induce activation and proliferation of pancreatic stellate cells (periacinar myofibroblasts), collagen deposition, and fibrosis (Fig. 19.5).

Autoimmune pancreatitis is a pathogenically distinct form of chronic pancreatitis that comes in three distinct forms, each with its own characteristic histopathology and clinical findings. Autoimmune pancreatitis type 1 (type 1 AIP) is associated with the presence of IgG4-secreting plasma cells in the pancreas and is one manifestation of IgG4-related disease (Chapter 6). In contrast, autoimmune pancreatitis type 2 (type 2 AIP) is restricted to the pancreas except for a subset of patients with ulcerative colitis. Both variants of autoimmune pancreatitis may mimic pancreatic carcinoma, including presentation as a “mass lesion” in the pancreatic head on imaging. Autoimmune pancreatitis, which responds to steroid therapy, is therefore important to distinguish from neoplasia. More recently, investigators have defined a third type of autoimmune pancreatitis, autoimmune pancreatitis type 3 (type 3 AIP), as a rare immune-related adverse event secondary to immune checkpoint inhibitor therapy that is often employed in the treatment of advanced malignancies. Type 3 AIP is more commonly observed with combination checkpoint inhibitor therapy than monotherapy, and most cases have mild symptoms.

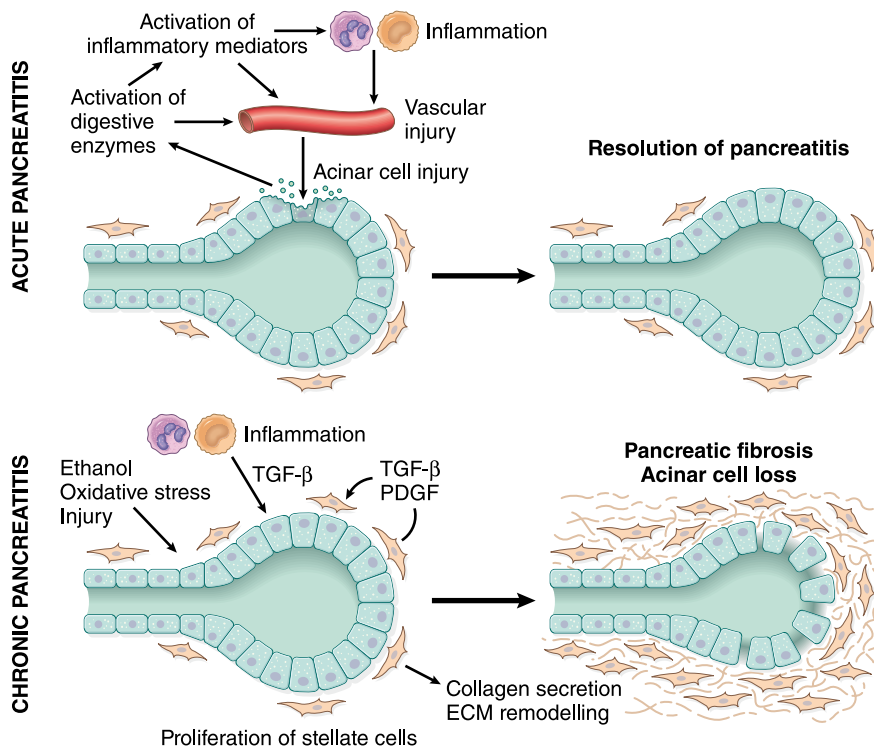


Fig. 19.5 Comparison of the mediators in acute and chronic pancreatitis. In acute pancreatitis, acinar injury results in release of digestive enzymes, leading to a cascade of events including activation of the clotting cascade, acute and chronic inflammation, vascular injury, and edema. In most patients, complete resolution of the acute injury occurs with restoration of acinar cell mass. In chronic pancreatitis, repeated episodes of acinar cell injury lead to the production of profibrogenic cytokines such as transforming growth factor- β ($TGF\beta$) and platelet-derived growth factor ($PDGF$), resulting in myofibroblast proliferation, collagen synthesis, and extracellular matrix (ECM) remodeling. Repeated injury leads to irreversible loss of acinar cell mass, fibrosis, and (exocrine) pancreatic insufficiency.

MORPHOLOGY

Chronic pancreatitis is characterized by parenchymal fibrosis, acinar atrophy and dropout, and variable ductal dilation (Fig. 19.6A). Grossly, the gland is hard, sometimes with visibly dilated ducts containing calcified concretions. These changes are typically accompanied by a chronic inflammatory infiltrate that surrounds lobules and ducts. Acinar loss is a constant feature, but there is usually relative sparing of the islets of Langerhans, which become embedded in the sclerotic tissue and may fuse and appear enlarged. Ductal epithelium may be atrophic, hyperplastic, or metaplastic (squamous). Chronic pancreatitis caused by alcohol abuse is characterized by ductal dilation and intraluminal protein plugs and calcifications (Fig. 19.6B). **Autoimmune pancreatitis** type 1 typically displays swirling or storiform fibrosis, obliterative inflammation of the veins (phlebitis), and a dense lymphoplasmacytic inflammation with abundant IgG4-secreting plasma cells (Fig. 19.6C). In contrast, autoimmune pancreatitis type 2 is characterized by a neutrophilic infiltrate within the ductal epithelium and lumen of medium-sized pancreatic ducts (granulocytic epithelial lesions). While lymphoplasmacytic infiltrates are seen in both variants, type 2 autoimmune pancreatitis lacks the abundant IgG4-secreting plasma cells that characterize type 1. The histology of type 3 autoimmune pancreatitis is unclear as most patients are diagnosed based on imaging studies and abnormal pancreas enzymes and resolve upon immune checkpoint inhibitor discontinuation.

Clinical Features

Chronic pancreatitis may follow multiple bouts of acute pancreatitis with repeated attacks of mild to moderately severe abdominal pain or persistent abdominal and back pain. Attacks may be precipitated by excessive alcohol consumption, overeating (which increases demand on the pancreas), or use of opiates and other drugs that increase the tone of sphincter of Oddi. In other patients the disease may be entirely silent until pancreatic insufficiency and diabetes mellitus develop due to destruction of the exocrine and endocrine pancreas.

Diagnosis of chronic pancreatitis requires a high degree of suspicion. During an attack there may be mild fever and mild-to-moderate eleva-

tions of serum amylase. In chronic disease, however, acinar cell dropout may be so great that this diagnostic clue is absent. Gallstone-induced obstruction may present with jaundice or elevated serum alkaline phosphatase. Visualization of calcifications within the pancreas by CT and ultrasonography can be very helpful. Weight loss (due to pancreatic exocrine insufficiency and malabsorption) and edema (secondary to hypoalbuminemia) may also be present.

Chronic pancreatitis is usually not an immediately life-threatening condition, but the long-term outlook is poor, with a 20- to 25-year mortality rate of 50%. Pancreatic exocrine insufficiency, chronic malabsorption, and diabetes mellitus can all lead to significant morbidity and contribute to mortality. In other patients, severe, chronic pain is a dominant problem. Pancreatic pseudocysts (described later) develop in about 10% of patients.

KEY CONCEPTS

Acute and Chronic Pancreatitis

- Acute pancreatitis is a form of reversible pancreatic parenchymal injury associated with inflammation.
- Acute pancreatitis may be caused by excessive alcohol intake, pancreatic duct obstruction (e.g., gallstones) and mutations in key genes (e.g., *PRSS1*, *SPINK1*, *CFTR*).
- The key underlying mechanism is inappropriate activation of digestive enzymes within the pancreas.
- Clinical features include acute abdominal pain, systemic inflammatory response syndrome, and elevated serum lipase and amylase levels.
- Chronic pancreatitis is characterized by irreversible injury of the pancreas leading to fibrosis, loss of pancreatic parenchyma, loss of exocrine and endocrine function, and pseudocyst development.
- Chronic pancreatitis is most often caused by:
 - Repeated bouts of acute pancreatitis
 - Chronic alcohol use

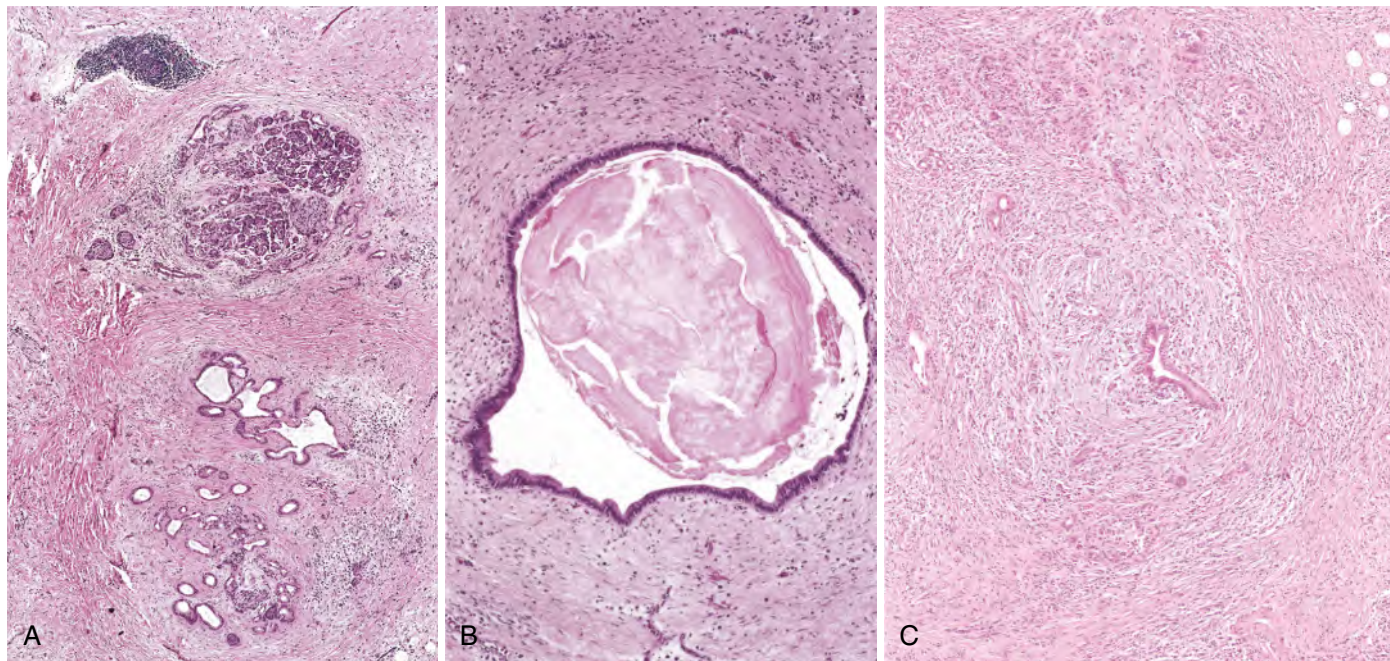


Fig. 19.6 Chronic pancreatitis. (A) Extensive fibrosis and atrophy has left only residual islets (*top*) and ducts (*bottom*), with a sprinkling of chronic inflammatory cells and a few islands of acinar tissue. (B) Higher power view demonstrating a dilated duct with inspissated eosinophilic ductal concretions in a person with alcohol-associated chronic pancreatitis. (C) Example of autoimmune pancreatitis with extensive lymphoplasmacytic infiltrates and storiform fibrotic stroma. In this case, an IgG4 stain confirmed abundant IgG4-expressing plasma cells. (C, Courtesy Aatur D. Singhi, University of Pittsburgh Medical Center.)

- Germline mutations in genes such as *CFTR* (the gene encoding the transporter that is defective in cystic fibrosis), particularly when combined with environmental stressors
- Chronic pancreatitis may be the manifestation of a systemic or localized autoimmune etiology.
- Clinical features include intermittent or persistent abdominal pain, intestinal malabsorption, and diabetes.

NONNEOPLASTIC CYSTS

A variety of cysts can arise in the pancreas. Most are nonneoplastic pseudocysts, but congenital cysts and neoplastic cysts also occur.

Congenital Cysts

Congenital cysts are unilocular, thin-walled cysts that likely result from anomalous pancreatic duct development. They range in size from microscopic to 5 cm in diameter and are lined by a uniform cuboidal epithelium or, if intracystic pressure is high, flattened and attenuated epithelial cells. A thin, fibrous capsule filled by clear serous fluid typifies congenital cysts that may be sporadic or part of inherited conditions such as *autosomal-dominant polycystic kidney disease* (Chapter 20) and *von Hippel-Lindau disease* (Chapter 28). Cysts in the kidney, liver, and pancreas frequently coexist in polycystic kidney disease. In von Hippel-Lindau disease, vascular neoplasms are found in the retina and cerebellum or brain stem in association with congenital cysts (and also neoplasms) in the pancreas, liver, and kidney.

Pseudocysts

Pseudocysts are formed when areas of intrapancreatic or peripancreatic hemorrhagic fat necrosis are walled off by fibrosis and granulation tissue. These lesions, which account for 35% of all pancreatic cysts, are

referred to as pseudocysts because they lack an epithelial lining. Pseudocysts typically arise following a bout of acute pancreatitis, particularly one superimposed on chronic alcoholic pancreatitis. Traumatic injury to the pancreas can also give rise to pseudocysts. While many pseudocysts resolve spontaneously, they may also become secondarily infected, and larger pseudocysts may compress or even erode into adjacent structures. However, not all pseudocysts arise after pancreatitis. Other etiologic factors include infection, neoplasms, and genetic disorders that can affect the pancreas.

MORPHOLOGY

Pseudocysts are usually solitary and may be situated within the pancreas or, more commonly, in the lesser omental sac or the retroperitoneum between the stomach and transverse colon or between the stomach and liver. They can even be subdiaphragmatic (Fig. 19.7A). Pseudocysts are lined by fibrous tissue and granulation tissue without a true epithelial lining (Fig. 19.7B) and range in size from 2 to 30 cm in diameter.

NEOPLASMS

A broad spectrum of exocrine neoplasms arises in the pancreas. These neoplasms may be cystic or solid; some are benign, while others are among the most lethal of all malignancies. Endocrine tumors also occur in the pancreas and are discussed in Chapter 24.

Cystic Neoplasms

Cystic neoplasms are diverse tumors that range from harmless benign cysts to lesions that are precursors to invasive, potentially lethal cancers. Only 5% to 15% of pancreatic cysts are neoplastic; conversely, fewer than 5% of pancreatic neoplasms are cystic. Serous cystic neoplasms

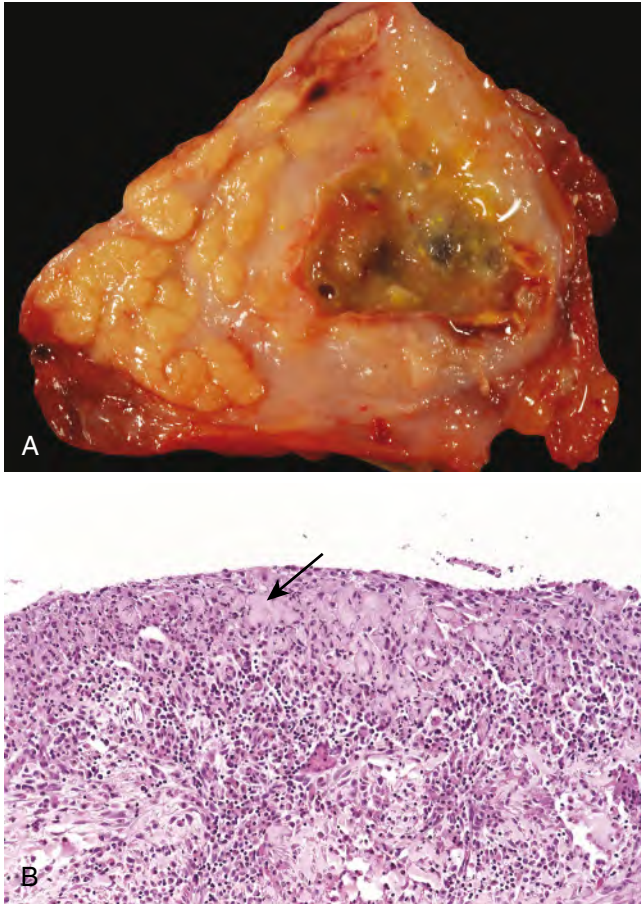


Fig. 19.7 Pancreatic pseudocyst. (A) Cross-section revealing a poorly defined cyst with a necrotic brown-black wall. (B) The cyst lacks a true epithelial lining and instead is lined by fibrin (arrow) and granulation tissue.

are almost always benign, whereas others, such as intraductal papillary mucinous neoplasms and mucinous cystic neoplasms, are precancerous. The genomic landscape of each type of cystic neoplasm is distinct, underscoring differences in pathogenesis and natural history.

Serous Cystic Neoplasms. Serous cystic neoplasms usually occur in the tail of the pancreas. The cysts are small (1 to 3 mm) and can be solitary, multiple, or present as a honeycomb of microcystic lesions. Serous cysts are lined by glycogen-rich cuboidal cells, and contain clear, thin, straw-colored fluid (Fig. 19.8). They account for 15% to 25% of all cystic neoplasms of the pancreas and are twice as common in females. Serous cystic neoplasms typically present in the sixth to seventh decade of life with nonspecific symptoms such as abdominal pain, but many are detected incidentally during imaging for another indication. Surgical resection is curative in most patients. Inactivation of the *VHL* tumor suppressor gene on chromosome 3p is the most common genetic abnormality in serous cystic neoplasms.

Mucinous Cystic Neoplasms. Close to 95% of mucinous cystic neoplasms arise in females and, in contrast to serous cystic neoplasms, they are precursors to invasive carcinomas. These neoplasms usually arise in the tail of the pancreas and present as painless, slow-growing masses. The cystic cavities are larger than those in serous cystic neoplasms and are filled with thick, tenacious mucin. The columnar mucin-producing epithelium that lines the cysts is associated with a dense stroma similar to ovarian stroma (Fig. 19.9). The latter frequently expresses estrogen

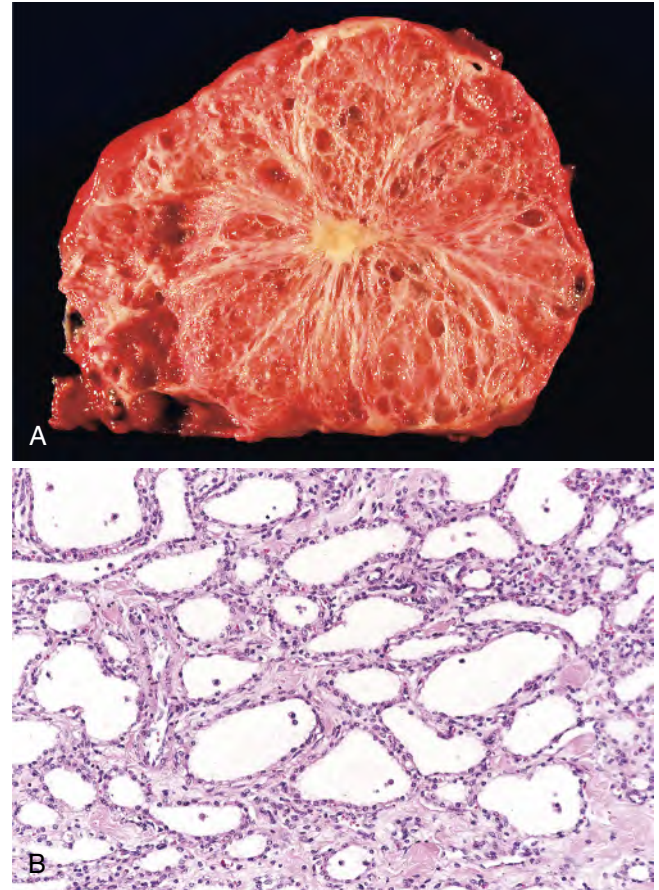


Fig. 19.8 Serous cystic neoplasm (serous cystadenoma). (A) Cross-section through a microcystic serous cystic neoplasm. Only a thin rim of normal pancreatic parenchyma remains. The cysts are relatively small and contain clear, straw-colored fluid. (B) The cysts are lined by cuboidal epithelium without atypia.

and progesterone receptors as well as other markers of the ovarian stroma, such as inhibin. Surgical resection is curative for noninvasive mucinous cystic neoplasms, but up to one-third harbor an invasive adenocarcinoma. Up to 50% of patients with an invasive adenocarcinoma arising in a mucinous cystic neoplasm will succumb to their disease within 5 years; therefore, early detection and treatment, before invasive cancer develops, is critical. Mucinous cystic neoplasms harbor oncogenic *KRAS* mutations in approximately half of cases, while mutations of *TP53* and *SMAD4* are typically observed in invasive neoplasms arising from these cysts. Loss-of-function mutations in *RNF43*, which encodes an E3 ubiquitin ligase that downregulates Wnt signaling, have also been described in up to one-third of mucinous cystic neoplasms. Similar *RNF43* mutations also occur in colorectal cancers.

Intraductal Papillary Mucinous Neoplasms. Intraductal papillary mucinous neoplasms (IPMNs) are mucin-producing neoplasms that involve the larger ducts of the pancreas. In contrast to mucinous cystic neoplasms, IPMNs arise more frequently in males and tend to involve the head of the pancreas. Up to 20% are multifocal. Two features are useful in distinguishing IPMNs from mucinous cystic neoplasms: (1) absence of the dense “ovarian” stroma seen in mucinous cystic neoplasms and (2) involvement of the main pancreatic duct or one of the branch ducts (Fig. 19.10). Just as with mucinous cystic neoplasms, IPMNs can progress to invasive cancer; early detection and treatment before progression to invasive cancer are thus critical. Similar to mucinous

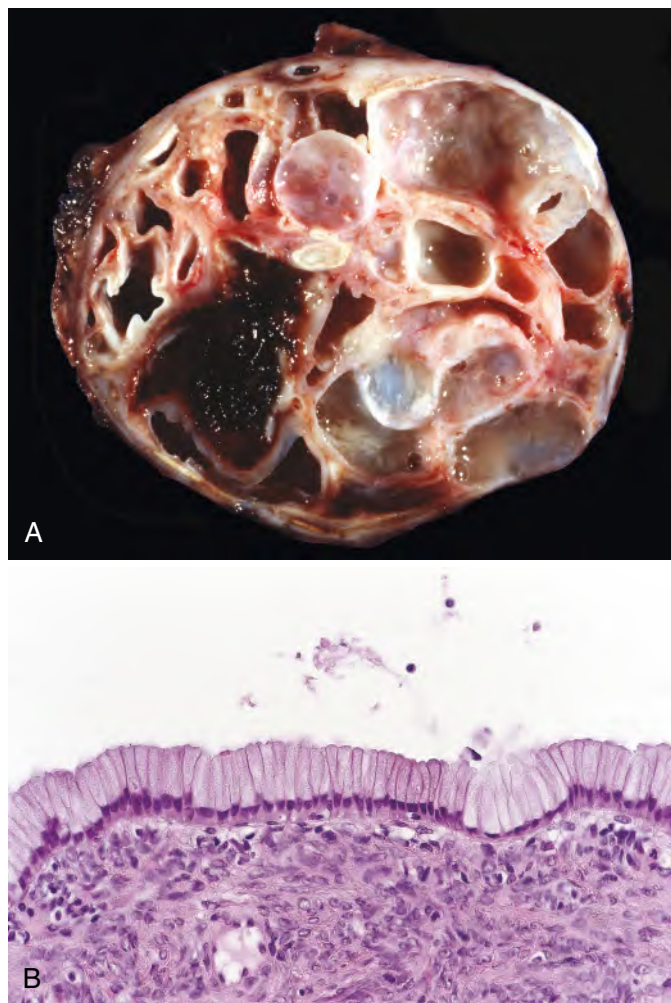


Fig. 19.9 Pancreatic mucinous cystic neoplasm with low-grade dysplasia. (A) Cross-section through a mucinous multiloculated cyst in the tail of the pancreas. The cysts are large and filled with tenacious mucin. (B) The cysts are lined by columnar mucinous epithelium with a dense “ovarian” stroma.

cystic neoplasms, intraductal mucinous papillary neoplasms harbor oncogenic *KRAS* mutations in approximately 80% of tumors and loss-of-function *RNF43* mutations in up to 50%. Further, *BRAF* alterations can occur in 10% of intraductal papillary mucinous neoplasms. *TP53* and *SMAD4* mutations typically occur only with the transition to invasive cancer. Notably, oncogenic mutations of *GNAS*, which encodes the alpha subunit of the stimulatory G protein, $G_{\alpha s}$, are present in approximately two-thirds of IPMNs but are not found in other pancreatic cysts.

Solid Pseudopapillary Neoplasms. The uncommon *solid-pseudopapillary neoplasm* occurs mainly in young females and due to its large size often presents with abdominal discomfort. As implied by the name, these well-circumscribed neoplasms have solid and cystic components, with the latter filled with hemorrhagic debris. The neoplastic cells grow in solid sheets or as pseudopapillary projections and often appear to be poorly cohesive. Activating mutations of *CTNNB1* (β -catenin) are present in nearly all these tumors, while mutations of *KRAS*, *RNF43*, *GNAS*, or *VHL* (characteristic of other pancreatic cystic neoplasms) are not observed. Surgical resection is the treatment of choice and proves curative in most patients.

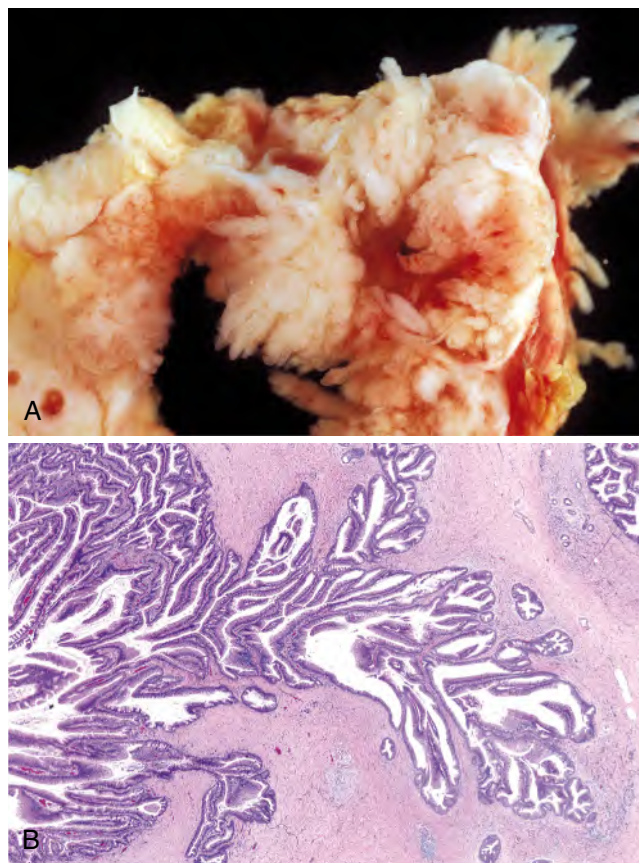


Fig. 19.10 Intraductal papillary mucinous neoplasm. (A) Cross-section through the head of the pancreas shows a prominent papillary neoplasm distending the main pancreatic duct. (B) The neoplasm involves the main pancreatic duct (*left*) and extends down into the smaller ducts and ductules (*right*).

KEY CONCEPTS

Cystic Neoplasms

- Virtually all serous cystic neoplasms are benign.
- Noninvasive intraductal papillary mucinous neoplasms and mucinous cystic neoplasms are almost always curable, but both may progress to invasive, potentially lethal carcinomas.
- Each of the major cystic neoplasms has a relatively specific mutational profile.

Pancreatic Carcinoma

Infiltrating ductal adenocarcinoma of the pancreas, the most common exocrine cancer of the pancreas, is the third leading cause of cancer deaths in the United States, trailing only lung and colon cancers, and has one of the highest mortality rates of any cancer. It was estimated that in 2023 pancreatic cancer would strike approximately 64,000 Americans, and approximately 50,000 will die from this disease, typically within a few months of diagnosis. The 5-year survival rate of pancreatic cancer has improved but remains a dismal 12%. The poor survival rates are largely driven by the overwhelming majority of patients (greater than 80%) presenting with cancers that are surgically unresectable and have spread into vital structures in the abdomen (e.g., major blood vessels), or to distant metastatic sites (e.g., liver).

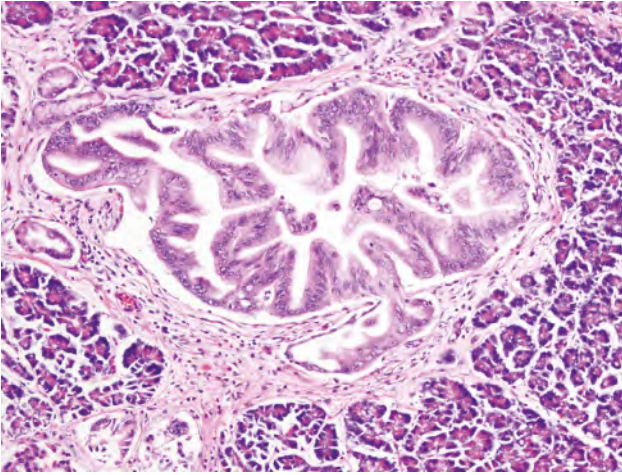


Fig. 19.11 High-grade pancreatic intraepithelial neoplasia involving a small pancreatic duct.

Precursors to Pancreatic Cancer

Invasive pancreatic cancers arise from noninvasive precursor lesions referred to as **pancreatic intraepithelial neoplasia (PanIN)** (Fig. 19.11). These lesions develop in small ducts and are often microscopic, although some, such as the two variants of cystic mucinous neoplasms described above, can be detected macroscopically. Overall, it is thought that greater than 85% of pancreatic cancers arise from PanIN (with the remainder arising from cystic lesions). This conclusion is supported by the following observations:

- The genetic and epigenetic alterations identified in PanIN are similar to those found in invasive cancers (described later).
- PanIN is often found in pancreatic parenchyma adjacent to infiltrating carcinoma. A corollary of this observation is that while isolated low-grade PanIN (formerly PanIN 1A, 1B, 2) lesions can be observed in the pancreata of elderly individuals, high-grade PanIN (formerly PanIN 3), also known as carcinoma in situ, is rarely observed in the absence of an invasive cancer.

- PanIN precedes the development of invasive cancer in genetically engineered mouse models of pancreatic cancer.
- Isolated case reports have documented individuals with PanIN who later developed invasive pancreatic cancer.

The epithelial cells in PanIN show dramatic telomere shortening, which may predispose these lesions to accumulate chromosomal abnormalities and progress to invasive carcinoma (Chapter 7). As previously mentioned, PanIN is separated into low-grade and high-grade based on morphologic features, which also tend to correlate with the stepwise accumulation of genetic abnormalities (Fig. 19.12).

Pathogenesis

As with other carcinomas, pancreatic cancer is the product of complementary mutations and epigenetic changes that alter the expression of oncogenes and tumor suppressor genes. The molecular basis of carcinogenesis and the functions of these genes were discussed in Chapter 7. The most common molecular alterations in pancreatic carcinogenesis are the following:

- **KRAS (chromosome 12p) is the most frequently altered oncogene in pancreatic cancer, with activating point mutations present in greater than 90% of cases.** The recent availability of small molecule inhibitors of mutant RAS proteins has ushered in an exciting new era in oncology, and multiple clinical trials are underway in patients with KRAS mutant pancreatic cancer.
- **CDKN2A (chromosome 9p)**, which encodes two tumor suppressors, p16/INK4a and ARF, is inactivated in 30% of pancreatic cancers through point mutations or homozygous deletions.
- **SMAD4 (chromosome 18q)**, which encodes a tumor suppressor involved in the TGF- β pathway, is inactivated in 55% of pancreatic cancers. *SMAD4* is only rarely inactivated in other cancer types.
- **TP53 (chromosome 17p)**, which encodes the tumor suppressor p53, is inactivated in 70% to 75% of pancreatic cancers.

Also seen are less common mutations that affect genes involved in DNA repair or regulation of chromatin structure (Table 19.3). In addition, other work has identified pathogenic epigenetic changes, which together with the aforementioned genetic changes produce several distinct molecular subtypes of pancreatic cancer.

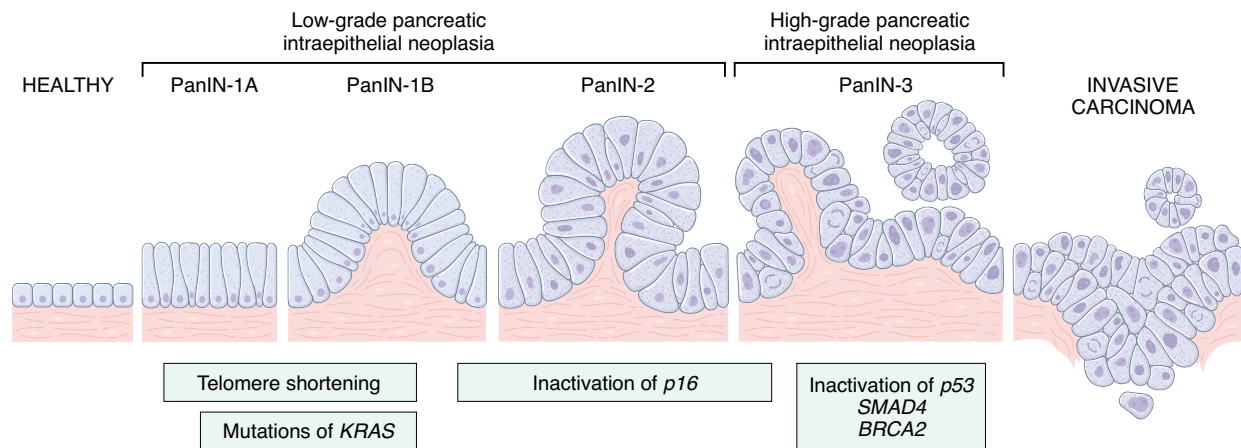


Fig. 19.12 Model for the progression from normal ducts (far left) through low- and high-grade pancreatic intraepithelial neoplasia (center) to invasive carcinoma (far right). Telomere shortening and mutations of the oncogene *KRAS* (PanIN-1) are the earliest discernible alterations in multistep progression, followed by inactivation of the *CDKN2A* tumor suppressor gene that encodes the cell-cycle regulator p16 (PanIN-2). Inactivation of the *TP53*, *SMAD4*, and *BRCA2* tumor suppressor genes occurs in high-grade pancreatic intraepithelial neoplasia (PanIN-3) that precedes invasive cancer. It is important to note that while there is a general temporal sequence of changes, the accumulation of multiple mutations is more important than the order of their occurrence. *PanIN*, Pancreatic intraepithelial neoplasia.

TABLE 19.3 Somatic Molecular Alterations in Pancreatic Ductal Adenocarcinoma

Gene	Chromosomal Region	Percentage of Cases With Genetic Alteration ^a	Gene Function
Oncogenes (Frequency Includes Point Mutations and Amplifications)			
<i>KRAS</i>	12p	>90%	GTP binding enzyme and growth factor signal transducer
<i>GATA6</i>	18q	9%	Transcription factor
<i>AKT2</i>	19q	6%	Growth factor signal transducer
<i>MYC</i>	8q	5%	Transcription factor
<i>GNAS</i>	20p	3%	Binds to adenylyl cyclase, increasing the conversion of ATP to cAMP
<i>BRAF</i>	7q	3%	Serine threonine kinase, regulates MAPK signaling
Tumor Suppressor Genes (Frequency Includes Point Mutations and Homozygous Deletions)			
<i>TP53</i>	17p	75%	Response to DNA damage
<i>SMAD4</i>	18q	55%	TGF- β pathway
<i>p16/CDKN2A</i>	9p	30%	Negative cell-cycle regulator
<i>ARID1A</i>	1p	6%	Chromatin regulator
<i>ATM</i>	11q	5%	DNA damage response
<i>MLL3/KMT2C</i>	7q	4%	Chromatin regulator
<i>BRCA2</i>	13q	4%	DNA damage response
<i>KDM6A</i>	Xp	3%	Chromatin regulator
<i>RNF43</i>	17q	3%	Negative regulator of the Wnt/ β -catenin pathway

^aFrequency of alterations abstracted from The Cancer Genome Atlas Research Network: Integrated Molecular Characterization of Pancreatic Ductal Adenocarcinoma. *Cancer Cell* 2017;32(2):185-203.e13.

cAMP, Cyclic adenosine monophosphate; GTP, guanosine triphosphate; MAPK, mitogen-activated protein kinase; TGF β , transforming growth factor- β .

- **DNA methylation (epigenetic) abnormalities.** DNA methylation abnormalities are widespread in pancreatic cancer. For example, promoter hypermethylation causes transcriptional silencing of tumor suppressor genes including *CDKN2A*. Conversely, promoter hypomethylation leads to overexpression of oncogenes such as *GATA6* and *BRD4*.
- **Transcriptomic profiles and pancreatic cancer subtypes.** Global analyses of gene expression have identified two distinct pancreatic cancer subtypes, termed basal-like and classic. Basal-like pancreatic cancers are highly aggressive, whereas the classic subtype has a somewhat more favorable prognosis. Molecular studies have found pancreatic cancers are composed of both subtypes in varying amounts and exhibit plasticity between these types in response to chemotherapy. How these subtypes respond differentially to specific therapies is an area of active investigation.

Epidemiology and Inheritance. Pancreatic cancer is primarily a disease of older adults, with 80% of cases occurring after age 60. The incidence is increased in African Americans, Japanese Americans, Native Hawaiian Islanders, and Ashkenazi Jewish people.

The strongest environmental influence is cigarette smoking, which doubles the risk of pancreatic cancer. Chronic pancreatitis is an additional risk factor that may reflect the association between chronic inflammation, tissue repair, and neoplasia. Pancreatic cancer risk is also greater in those with visceral obesity and high body mass index. Diabetes mellitus is a modest risk factor. Conversely, new-onset diabetes may be the first indication of occult pancreatic cancer in older patients; abnormal glucose tolerance or diabetes is present in as many as half of pancreatic cancer patients up to 2 years before clinical signs of cancer appear. Thus, the appearance of new-onset diabetes may present an opportunity for earlier diagnosis of an occult pancreatic cancer.

Approximately 10% of patients with pancreatic cancer have a deleterious germline mutation in a cancer predisposition gene (Table 19.4)

or report one or more first-degree relatives with pancreatic cancer. Germline mutations of *BRCA2* and *ATM*, both of which encode DNA repair proteins, are the two most common known causes of familial pancreatic cancer; germline mutations in other genes within the same DNA repair family such as *BRCA2* (*BRCA1*, *PALB2*) are also associated with variably elevated lifetime risk, which is further increased if there are first-degree relatives with pancreatic cancer. With the availability of therapeutic agents that specifically target cancers with DNA repair defects, germline testing is now recommended for all patients with pancreatic cancer. Identification of germline mutations in the index case also provides an opportunity for testing otherwise asymptomatic first-degree relatives who may harbor the identical alteration, and can then enroll in surveillance programs for early detection. Patients with defects in DNA mismatch repair genes who have hereditary nonpolyposis colorectal cancer are also at increased risk for pancreatic cancer. Identification of microsatellite instability, the molecular signature of DNA mismatch repair in pancreatic cancer, is also therapeutically important, as such cancers are more likely to respond to immune checkpoint therapy.

MORPHOLOGY

Approximately 60% of pancreatic cancers arise in the head, 15% in the body, and 5% in the tail; in 20% of cases the entire pancreas is involved.

Most pancreatic neoplasms are adenocarcinomas that recapitulate normal ductal epithelium by forming glands and secreting mucin. Two features are characteristic: (1) these cancers are **highly invasive**, often extending into peripancreatic tissues; and (2) they elicit an intense **desmoplastic response** that results in the deposition of dense collagen. As a result of these characteristics, most pancreatic cancers are hard, stellate, gray-white, poorly defined masses (Fig. 19.13A).

Most carcinomas of the head of the pancreas obstruct the distal common bile duct resulting in distention of the biliary tree in about 50% of patients. In contrast, carcinomas of the pancreatic body and tail do not impinge on the biliary tract and may be large and widely disseminated at diagnosis.

TABLE 19.4 Inherited Predisposition to Pancreatic Cancer

Disorder	Gene	Increased Risk of Pancreatic Cancer (Fold)	Risk of Pancreatic Cancer by Age 70 Years (%)
Peutz-Jeghers syndrome	<i>STK11</i>	130	30–60
Hereditary pancreatitis	<i>PRSS1</i> , <i>SPINK1</i>	50–80	25–40
Familial atypical multiple-mole melanoma syndrome	<i>CDKN2A</i>	20–35	10–17
Strong family history (3 or more relatives with pancreatic cancer)	Unknown	14–32	8–16
Hereditary breast and ovarian cancer	Multiple, including <i>BRCA1</i> , <i>BRCA2</i> , <i>PALB2</i> , <i>ATM</i>	4–10	5
Lynch syndrome (also known as hereditary non-polyposis colorectal cancer)	Multiple, including <i>MLH1</i> , <i>MSH2</i> , <i>PMS2</i>	8–10	4

Pancreatic cancers tend to grow along nerves and invade blood vessels and the retroperitoneum. The spleen, adrenals, transverse colon, and stomach are often involved by direct invasion, and peripancreatic, gastric, mesenteric, omental, and portohepatic lymph nodes are frequently involved. Perineural, lymphatic, and large vessel invasion are common. Distant metastases are principally to the liver and lungs.

Microscopically, carcinomas of the head, body, and tail of the pancreas are indistinguishable. They form abortive tubular structures or cell clusters with an aggressive, deeply infiltrative growth pattern (Fig. 19.13B). The poorly formed malignant glands are lined by pleomorphic cuboidal-to-columnar epithelial cells. It can, however, occasionally be difficult to discriminate between well-differentiated carcinomas and benign glands with epithelial atypia and stromal fibrosis that are common in chronic pancreatitis. The intense stromal fibrosis, termed **desmoplasia**, elicited by pancreatic cancers can also interfere with diagnostic interpretation of needle biopsies, regardless of tumor differentiation, as much of the tumor mass is nonneoplastic.

Less common morphologic variants of pancreatic cancer include adenosquamous carcinomas, colloid carcinoma, hepatoid carcinoma, medullary carcinoma, signet-ring cell carcinoma, undifferentiated carcinoma, and undifferentiated carcinoma with osteoclast-like giant cells.

Clinical Features

Carcinomas of the pancreas typically remain silent until they cause obstruction or invade adjacent structures. Pain is usually the first symptom, and by the time pain appears these cancers are usually beyond cure. Obstructive jaundice is associated with most cases involving the pancreatic head, as these tend to block the common bile duct; this is exemplified by the *Courvoisier sign*, a palpably enlarged, nontender gallbladder with mild painless jaundice. As previously mentioned, new-onset diabetes may be a sign of an underlying pancreatic cancer that has not yet clinically manifested itself. In contrast, weight loss, anorexia, generalized malaise, and weakness are often signs of advanced disease. Profound loss of skeletal muscle mass and visceral fat (cancer-associated *cachexia*) is common in advanced disease and contributes to the overall systemic deterioration observed in many patients. Migratory thrombophlebitis, known as the *Trousseau sign*, occurs in about 10% of patients and is attributable to the elaboration of platelet-activating factors and procoagulants from the carcinoma or its necrotic products (Chapter 4). Sadly, Armand Trousseau (1801–1867, physician at Hotel Dieu, Paris), for whom this sign is named, correctly suspected that he had carcinoma when he developed spontaneously appearing and disappearing (migratory) thromboses.

Survival after diagnosis of advanced pancreatic carcinoma is typically short. Greater than 80% of pancreatic cancers are unresectable at the time of diagnosis due to invasion of vessels and other structures

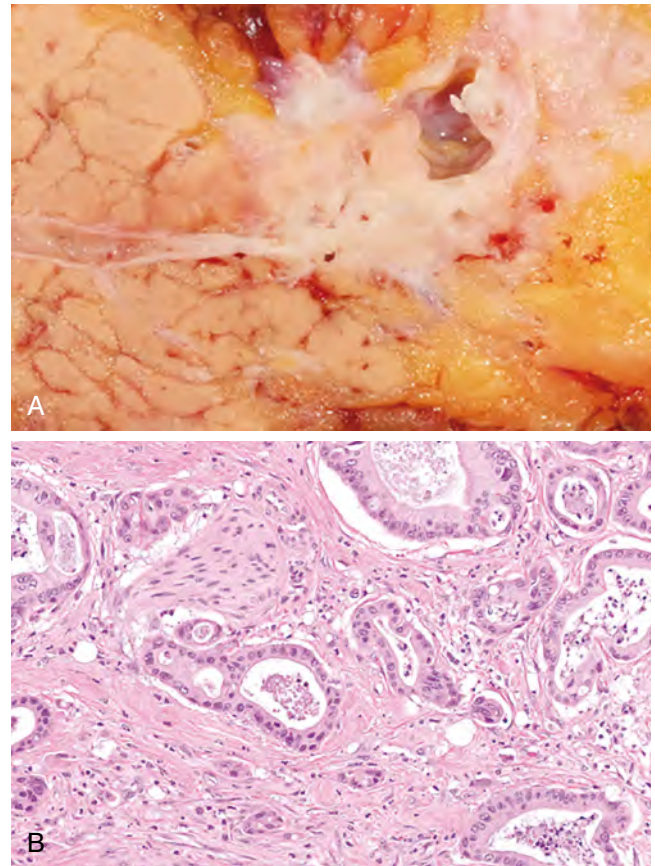


Fig. 19.13 Carcinoma of the pancreas. (A) A cross-section through the tail of the pancreas exhibiting normal pancreatic parenchyma and a normal pancreatic duct (*left*) and a white mass centered on the duct (*right*). (B) Moderately to poorly formed glands composed of malignant epithelial cells are dispersed within densely fibrotic stroma and invading a nearby nerve; some inflammatory cells are also present.

or distant metastases. In contrast, patients who undergo successful resection are increasingly surviving longer (some greater than 5 years), underscoring the critical importance of early detection. To facilitate diagnosis when pancreatic cancers are resectable, many have sought tests that could be used for early detection. Although serum levels of secreted antigens (e.g., carcinoembryonic antigen and CA19-9 antigen) or circulating tumor DNA (ctDNA) shed from tumor cells are often elevated in individuals with pancreatic cancer and can be useful in following individual patient's response to treatment, these markers lack the sensitivity needed for diagnosing most

early-stage cancers, and the specificity for application to the general population at large. Similarly, imaging techniques such as endoscopic ultrasonography and CT can be used to establish the diagnosis in patients presenting with suspicious symptoms, but are not useful as screening tests in the general asymptomatic population. Thus, the US Screening and Prevention Task Force (USPSTF) does not currently recommend screening the general population for pancreatic cancer due to the overarching concern for false positives and potential harm. However, longitudinal surveillance is recommended for individuals who harbor deleterious germline mutations (see Table 19.4) that place them at increased lifetime risk for pancreatic cancer.

KEY CONCEPTS

Pancreatic Carcinoma

- Pancreatic cancer is one of the most aggressive of solid cancers.
- Cigarette smoking is a significant cause of pancreatic cancer.
- Cancer-causing germline mutations are present in 10% of patients.
- Invasive pancreatic cancer arises from histologically well-defined precursor lesions, the most common of which is pancreatic intraepithelial neoplasia (PanIN).
- Ductal adenocarcinomas are highly invasive and elicit an intense desmoplastic response.
- The genes most frequently mutated or otherwise altered in pancreatic cancer include *KRAS*, *p16/CDKN2A*, *TP53*, and *SMAD4*; transcriptional profiles can be used to define the highly aggressive basal-like and somewhat less aggressive classic subtypes.
- Patients often present with abdominal pain and weight loss, sometimes accompanied by jaundice and deep vein thrombosis. New-onset diabetes is detected in up to half of cases.

Acinar Cell Carcinoma

Like normal acinar cells, acinar cell carcinomas form zymogen granules and produce exocrine enzymes such as trypsin and lipase. Acinar cell carcinoma of pancreas accounts for 1% to 2% of pancreatic tumors in adults. Up to 15% of those with acinar cell carcinoma develop metastatic fat necrosis syndrome due to lipase release into the circulation. Acinar cell carcinomas demonstrate aberrant Wnt pathway activation, due to either loss-of-function mutations of the *APC* tumor suppressor gene or activating point mutations of *CTNNB1*, which encodes beta-catenin.

Pancreatoblastoma

Pancreatoblastomas are rare neoplasms that occur primarily in children 1 to 15 years of age. These neoplasms have a distinct microscopic appearance consisting of squamous islands admixed with acinar cells. They are malignant, but survival is better than with pancreatic ductal adenocarcinomas. Like acinar cell carcinoma, pancreatoblastomas frequently harbor mutations that activate the Wnt signaling pathway.

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The Kidney

Astrid Weins and Anthony Chang

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